Language Query-Based Transformer With Multiscale Cross-Modal Alignment for Visual Grounding on Remote Sensing Images

Meng Lan, Fu Rong, Hongzan Jiao, *Member, IEEE*, Zhi Gao, *Member, IEEE*, and Lefei Zhang, *Senior Member, IEEE*

***Abstract*— Visual grounding for remote sensing images (RSVG) aims to localize the referred objects in the remote sensing (RS) images according to a language expression. Existing methods tend to align visual and text features followed by concatenation and then employ a fusion Transformer to learn a token representation for final target localization. However, simple fusion Transformer structure fails to sufficiently learn the location representation of referred object from the multimodal features. Inspired by the detection transformer (DETR), in this article, we propose a novel language query-based Transformer framework for RSVG termed LQVG. Specifically, we adopt the extracted sentence-level text features as the queries, called language queries, to retrieve and aggregate representation information of the referred object from the multiscale visual features in the Transformer decoder. The language queries are then converted into object embeddings for final coordinate prediction of referred object. Besides, a multiscale cross-modal alignment (MSCMA) module is devised before the multimodal Transformer to enhance the semantic correlation between the visual and text features, thus facilitating the cross-modal decoding process to generate more precise object representation. Moreover, a new RSVG dataset named RSVG-HR is built to evaluate the performance of the RSVG approaches on very high-resolution RS images with inconspicuous objects. Experimental results on two benchmark datasets demonstrate that our proposed method significantly surpasses the comparison methods and achieves state-of-the-art performance. The dataset and code are available at https://github.com/LANMNG/LQVG.**

***Index Terms*— Detection transformer (DETR), multimodal learning, remote sensing (RS), visual grounding.**

I. Introduction

VISUAL grounding on remote sensing images (RSVG) is an emerging task for remote sensing (RS) image processing, which aims to automatically localize the objects described by a natural language expression in RS images. Compared with traditional object detection task [1], [2], [3] that detects all the objects seen in the training set, visual grounding can localize the specified targets more flexibly. Moreover, RSVG allows nonspecialized users to easily retrieve and access target information in RS images in a user-friendly manner, which has wide application prospects in scenarios such as military target localization and recognition, disaster response, military intelligence analysis, and agricultural monitoring [4], [5], [6].

Since RSVG task is a new topic in the community, fewer works are explored in this field. Sun et al. [7] first introduced the concept of visual grounding on RS images and proposed a RSVG dataset and the GeoVG method. The GeoVG framework is mainly composed of a language encoder, an image encoder, and a fusion module. The language encoder uses a numerical context module to represent a complex expression as a geospatial relation graph, and the image encoder introduces an adaptive region attention module to distill key visual content. Finally, the fusion module adopts a self-attention network to fuse the text and image features for visual grounding. The proposed RSVG dataset is built on top of large-scale RS images, where each image covers an area ranging from 0.06 to 25 km^2 with complex background and inconspicuous objects. This dataset can well validate the performance of the RSVG methods on high-resolution RS images, however, the language annotations of the dataset essentially rely on the distance and area attributes of the target objects, which leads to the incompatibility of this dataset with the conventional visual grounding methods. Furthermore, it is usually not easy to obtain accurate location and area information of the target, thus resulting in limitations of the RSVG dataset in practical applications. Recently, Zhan et al. [8] proposed a new RSVG approach called MGVLF as well as a new RSVG dataset, i.e., DIOR-RSVG, based on the target detection dataset DIOR [9]. MGVLF uses a convolutional neural network (CNN) to extract multiscale visual features and the BERT model [10] to obtain word- and sentence-level text embeddings, and then the multigranularity visual language fusion module is applied on them to enable the visual representation to focus on the valid regions. Finally, a transformer-based multimodal fusion module is employed to produce a learnable token for object localization. The proposed DIOR-RSVG dataset is a new

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Meng Lan, Fu Rong, and Lefei Zhang are with the Institute of Artificial Intelligence, School of Computer Science, Wuhan University, Wuhan 430072, China, and also with Hubei Luojia Laboratory, Wuhan 430079, China (e-mail: menglan@whu.edu.cn; furong@whu.edu.cn; zhanglefei@whu.edu.cn).

Hongzan Jiao is with the School of Urban Design, Wuhan University, Wuhan 430072, China (e-mail: jiaohongzan@whu.edu.cn).

Zhi Gao is with the School of Remote Sensing and Information Engineering, Wuhan University, Wuhan 430079, China (e-mail: gaozhinus@gmail.com).

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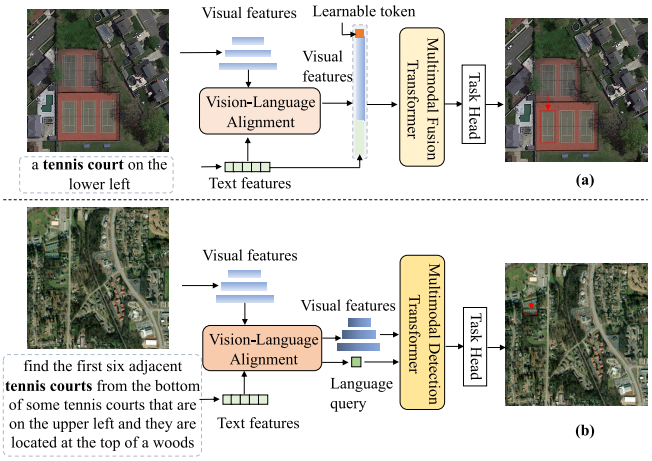


Fig. 1. Comparison of RSVG frameworks. (a) Existing fusion Transformer-based RSVG methods. The input RS image–text pair is from DIOR-RSVG dataset. (b) Our proposed language query-based RSVG model LQVG. The input RS image–text pair is from our proposed RSVG-HR dataset.

large-scale benchmark for RSVG task and fully advances the research of RSVG. However, the referred objects in the RS images of DIOR-RSVG dataset are highly salient, as shown in Fig. 1(a), which may decrease the challenge of cross-modal visual grounding and make it difficult to meet the practical requirements of various RS image analysis tasks. This drives us to figure out whether RSVG algorithms can still be effective on very high-resolution RS images that contain inconspicuous objects using simple and common language descriptions.

On the basic of the RSVG dataset in [7], we construct the RSVG-HR dataset by sampling a subset from the RSVG dataset and reannotating the sampled high-resolution RS images with conventional language descriptions using a new annotation protocol. RSVG-HR dataset has seven categories of objects and contains 2650 image–text pairs for training and test. Compared with the RS images in the DIOR-RSVG dataset, RS images in RSVG-HR have higher resolution and cover a larger area with complex background and inconspicuous objects. Moreover, the language annotations of RSVG-HR conform to the annotation conventions of traditional visual grounding algorithms, thus RSVG-HR is compatible with other RSVG datasets and could be used to evaluate the performance of RSVG models that do not need specialized modifications.

Previous RSVG methods [7], [8] follow the architectural strategy in Fig. 1(a) when designing their models. In the strategy, the image encoder and text encoder extract the visual and text features followed with various vision-language alignment module to refine the visual feature, which is then concatenated with the text features and a learnable token. The concatenated features are fed into a multimodal fusion Transformer where the intramodal and intermodal representation information are gathered into the learnable token through the self-attention mechanism. Eventually, the learnable token is sent to the localization module to regress the box coordinate. Nevertheless, there are two potential issues in this design paradigm. *First*, as the referred object always occupies a small spatial place in the complex large-scale RS image, the target visual representation is limited in the sequence of concatenated cross-modal features, and this may make it hard for the learnable token to gather target information through the self-attention mechanism. *Second*, only one learnable token may be not enough to learn the positional relationships and context features of the referred object in complex RS images, which in turn may lead to inaccurate localization.

Inspired by the detection Transformer (DETR) architecture [1], in this article, we introduce the query-based Transformer architecture and propose a novel language query-based multimodal Transformer framework for RSVG task, termed LQVG. As shown in Fig. 1(b), LQVG mainly consists of three parts, namely the feature extraction, the multiscale cross-modal alignment (MSCMA) module and the multimodal DETR. The image and text encoders take the RS image and the corresponding description as input and generate the multiscale visual features and the word-level text features (text features), respectively. Then, the visual and text features are fed into the vision-language module, i.e., our proposed MSCMA module, to enhance the semantic correlation between the cross-modal features. The MSCMA module delivers the refined multiscale visual features and the refined text features. Subsequently, the text features are converted to sentence-level text feature (sentence feature) through a projection layer which is input into the multimodal DETR as the language queries along with the refined multiscale visual features. After the cross-modal interaction in the Transformer, the output object embeddings are sent to the task heads to predict the coordinate of object bounding box. Notably, in our model, the language queries directly interact with the visual features through the cross-attention layers, rather than as a whole to acquire the object embeddings via the self-attention. This allows all the queries use the language expression as guidance to find the referred objects only. Moreover, there are multiple queries embeddings that are initialized with the sentence feature, and they can comprehensively learn the contextual information of referred object in the complex RS image, and then we select the object embeddings with the strongest representation to predict the accurate target coordinate.

In summary, we conclude our contributions as follows.

1) We propose a novel language query-based multimodal DETR framework for RSVG task, in which the sentence feature serves as the queries to retrieve and aggregate target representation from the multiscale visual features and the learned object embeddings are used to predict the coordinate of referred object.
2) We devise a MSCMA module to enhance the semantic correlation between the visual and text features, and facilitate the cross-modal decoding process in the subsequent Transformer decoder for better performance.
3) We construct a new high-resolution RS images-based dataset for RSVG task, termed RSVG-HR, and experimental results on the RSVG-HR and DIOR-RSVG datasets demonstrate that our proposed LQVG achieves state-of-the-art performance on both RS images with

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salient objects and high-resolution RS images with inconspicuous objects.

The rest of the article is organized as follows. Section II briefly reviews the related work. In Section III, we elaborate our proposed LQVG model. Section IV describes the details of the RSVG-HR dataset and provides extensive experimental results. In the end, we conclude the article in Section V.

II. Related Work

In this section, we provide a comprehensive review of methods pertaining to visual grounding for both natural and RS images.

A. Visual Grounding on Natural Images

Existing methods for visual grounding on natural images can be categorized into three main groups: two-stage, one-stage, and transformer-based methods.

The major concept of two-stage visual grounding methods [11], [12], [13], [14], [15] is to first use a pretrained object detector to generate candidate region proposals, and then select the most matching object with the textual expression as the detection result. The primary focus of these approaches lies in the second stage, which involves addressing visual grounding as text-region matching. Hu et al. [11] proposed the Spatial Context Recurrent ConvNet (SCRC) model, which takes the text query, candidate regions, their spatial configurations, and global context as input, and produces scores for each candidate region. Zhang et al. [16] introduced a variational Bayesian model that exploits the reciprocal relation between the referent and context for the complex context modeling and thereby reduces the search space of context. A cross-modal relationship extractor is proposed in [17]. It utilizes the cross-modal attention mechanism to adaptively emphasize objects and semantic relationships associated with the given expression. However, these approaches are heavily dependent on the performance of the region proposal networks. Furthermore, in RS images, the abundance of acquired objects necessitates the second stage to filter a substantial number of candidate regions, leading to potentially performance degradation and long inference time.

The one-stage methods prevent predicting the location of the referring object by generating a large number of candidate proposals, which is more efficient and avoids the error accumulation of the multistage framework [18], [19]. For instance, FAOA [20] encodes language referents as text feature embeddings and fuses them with the YOLOv3 detector [21], which ultimately selects the target with the highest confidence score as the prediction. RCCF [22] considers the visual grounding task as a correlation filtering process. It maps the text features into filtering kernels and applies filtration to image features. Yang et al. [23] proposed a recursive subquery construction network to address the query modeling deficiency, which reasons between image and query for multiple rounds and reduces the referring ambiguity step by step.

In recent years, Transformer [24] has been widely used in various computer vision task [25], [26], [27], [28], including the visual grounding task [29], [30], [31] because of its abilities of long-range relationship modeling and handling the multimodal data. TransVG [31] concatenates the visual and text features as well as an additional memory token, and then put them into the Transformer, where the memory token is used to aggregate target information for coordinate prediction. In order to retain the advantages of the pretrained vision transformer, Deng et al. [32] proposed the TransVG++ that feeds the language features into the vision encoder and performs the multimodal fusion directly in the vision transformer. QRNet [33] proposed a query-aware dynamic attention mechanism to dynamically compute query-dependent spatial and channel attentions for refining visual features. VLTVG [34] enhances visual features with a visual-language verification module and aggregate visual context using a language-guided context encoder.

B. Visual Grounding on RS Images

Compared to visual grounding algorithms designed for natural images, the field of visual grounding on RS images has a relatively delayed inception and currently lacks sufficient exploration and study. In contrast to natural images, target objects in RS images may not be visually distinguishable from the image background in terms of the scale and color of the object and so on. Moreover, RS images often contain multiple similar objects, which requires the visual model to pay more attention to the correlations between objects and be capable of accurately understanding the expressive intent of language expressions. Sun et al. [7] first proposed the RSVG task and collected a dataset for the task. They designed the GeoVG model to capture the local correlation of the objects in the RS images and better understand the spatial information of the objects. However, the lack of compatibility in the language annotations of the RSVG dataset limits its practice application. Zhan et al. [8] also presented a new dataset DIOR-RSVG along with the MGVLF framework for the RSVG task. The DIOR-RSVG dataset was constructed by annotating the language expressions for the object detection dataset DIOR [9]. Considering the difference between natural images and RS images, the MGVLF model devised a multigranularity cross-modal module based on the TransVG [31] model to learn more discriminative visual representation by interacting the multiscale visual features with multigranularity text features. Although these methods have achieved promising performance, we argue that the simple concatenation of the visual and text features may not be a good fusion manner for the only one learnable object token to learn the location information of referred object from the cross-modal features. Despite the robust multimodal understanding capabilities demonstrated by previous methods [35], [36], [37] in other RS multimodal tasks, there is an urgent need for the development of excellent cross-modal interaction mechanisms in the RSVG task. In this article, we propose a novel language query-based multimodal Transformer that directly uses sentence feature to initialize multiple object queries to retrieve and aggregate the target representation, and generate the object embeddings for location prediction.

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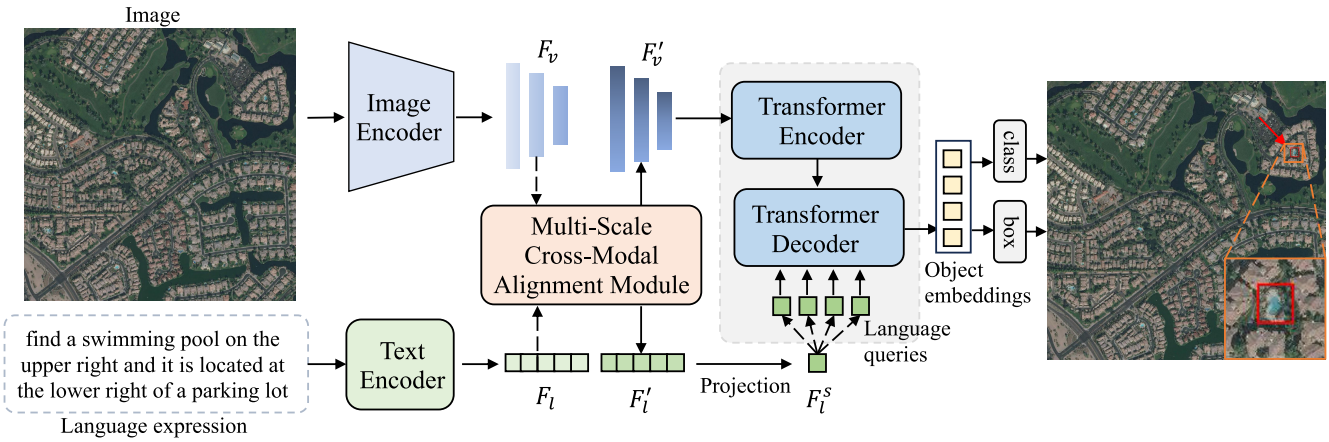


Fig. 2. Overview of our proposed LQVG framework. The model first aligns the multiscale visual features and the text features through the multiscale cross-modal alignment module and then takes the sentence feature as language queries to retrieve and aggregate the referred object information from the multiscale visual feature in the multimodal Transformer. The Transformer produces the object embeddings for the final object coordinate prediction.

III. Method

A. Overview

The overview of our proposed LQVG is illustrated in Fig. 2. It mainly consists of three parts: the image and text encoders, the MSCMA module, and the multimodal DETR. During inference, given an RS image $I \in R^{H\times W\times 3}$ and the corresponding language expression of the referred object $\mathcal{E} = \{e_l\}_{l=1}^{L}$ with L words, the image and text encoders first extract the multiscale visual features of the input RS image and the text features of the language expression, which are then fed into the MSCMA module to align the semantics of the multimodal features and produce the refined visual and text features. The text features are subsequently converted into the sentence feature via a projection layer. After that, the multimodal DETR encoder takes the refined multiscale visual features as input and its outputs along with the sentence feature are fed into the Transformer decoder, where the sentence feature serves as the language queries to decode the object information from the image features and generates the object embeddings. Finally, the object embeddings go through the class head and the box head to predict the confidence score and the bounding box coordinate.

B. Feature Extraction

1) Image Encoder: For the input RS image, we adopt the widely used ResNet [38] as the image encoder to extract the multilevel visual features $F_v = \{F_v^i\}_{i=1}^{4}$, where F_i denotes the i-th level features. Specifically, for the multilevel visual features F_v, the first three-level features are the last three stage features of the image encoder with spatial strides of $\{8, 16, 32\}$, and the last-level feature is obtained by downsampling the 32-stride feature using a convolutional layer with stride 2, thus F_v is the four-level multiscale features with strides of $\{8, 16, 32, 64\}$.

2) Text Encoder: For the input language expression, the text encoder generates the text features $F_l \in R^{L\times C}$ that contain the feature embedding of each word in the language expression. Since text features are semantically rich and contain more object information, they are more suitable for fine-grained cross-modal interactions with multiscale visual features to align the target semantics in the multimodal features.

C. MSCMA Module

Since the high-level sentence-level text feature will serve as the object queries to interact with the multiscale visual features in the multimodal decoder, the weak semantic correlation between visual and text features before multimodal Transformer may exacerbate the challenge of accurately decoding the object information from visual features. Therefore, it is crucial to enhance the semantic correlation of the cross-modal features so as to facilitate the cross-modal decoding process in the multimodal Transformer for accurate target object localization.

Different from the visual language fusion manner designed in the previous MGVLF [8], which performs the interaction between the single-level visual features and the concatenated multiscale visual features and multigranularity text embeddings to refine the visual features, in this article, we devise the MSCMA module, which directly implements the cross-modal interaction between the multiscale visual features and text features, to align the semantics and produce both refined multiscale visual features and refined text features before the multimodal DETR.

As shown in Fig. 3, the MSCMA module consists of two submodules, namely language–vision interaction (LVI) submodule and vision–language interaction (VLI) submodule. Both submodules take the text feature F_l and multiscale visual features F_v as input, and the LVI submodule outputs the refined text features F_l' while the VLI submodule produces the refined multiscale visual features F_v'. We will describe these two submodules in detail in the following part.

1) Language–Vision Interaction Submodule: LVI submodule progressively performs cross-modal interactions between the text features and the different level visual features, where the visual features serve as the guidance to refine the text features and build the correlation between the visual and text features. For simplicity, we take the example of three-level features here. As shown in Fig. 3(a), the text features F_l and the

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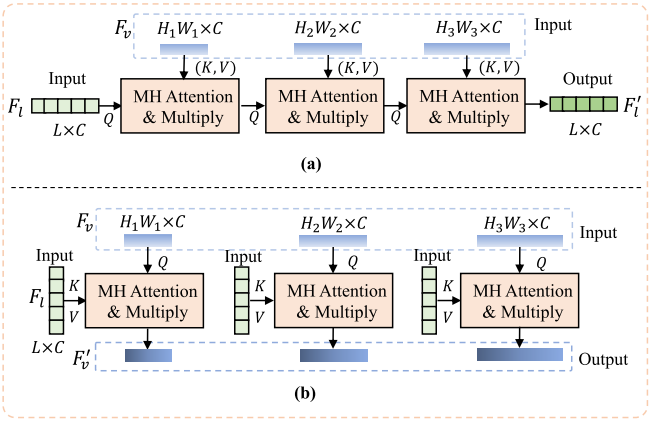


Fig. 3. Illustration of the proposed MSCMA module. MH Attention denotes the multihead attention layer. Multiply denotes the element-wise multiplication operation. (a) Language–Vision interaction submodule. (b) Vision–Language interaction submodule.

first level visual features F_v^3 first realize the interaction through a multihead attention layer and a multiplication layer [28]. The output refined text features then iteratively interact with the rest visual features until the text features interact with all levels of the multiscale visual features.

The multihead attention layer adopts the cross-attention mechanism to accomplish the cross-modal interaction between the text and visual embeddings, where the text embeddings $F_l \in R^{L\times C}$ serves as the query (Q) and the single-level visual features $F_v^i \in R^{H_iW_i\times C}$ is used as the key-value pair (K, V). Here, H_i and W_i are the height and width of the ith level of features, and C is the channel number. For the ith iteration, the attention layer first obtains the cross-modal similarity matrix $A_{lv}^i \in R^{L\times TH_iW_i}$ by computing the similarity between each word embedding and each pixel embedding as follows:

$$A_{lv}^i = \text{Softmax}\left(\frac{F_l^i W^Q \cdot \left(F_v^i W^K\right)^T}{\sqrt{d_k}}\right) \tag{1}$$

where $W^Q and W^K \in R^{C\times d_k}$ are learnable linear projections. Then, we use A_{lv}^i to aggregate the language-related object information in the visual features and multiply with the input text features F_l^i to obtain the refined text embeddings F_l^{i+1}

$$F_l^{i+1} = \left(A_{lv}^i F_v^i W^V\right) \cdot F_l^i \tag{2}$$

where W^v is a learnable linear projection. To obtain positional information, a fixed 2-D sinusoidal positional encoding is added to the visual features before the cross-attention process. The raw text features F_l is F_l^1 in the first iteration.

After the last iteration of the cross-modal interaction, the final refined text features F_l' are converted to the sentence feature $F_l^s \in \mathbb{R}^{1\times C}$ via a projection layer. In the projection layer, the [CLS] embedding is first selected from the text embeddings and then a linear layer followed with a Tanh layer is applied to the [CLS] embedding to generate the sentence feature.

2) Vision–Language Interaction Submodule: VLI submodule has similar submodule structure and input with the LVI submodule, while the VLI submodule runs a different procedure. As presented in Fig. 3(b), each level of the original multiscale visual features F_v^i interacts with the raw text embeddings F_l individually via the multihead attention layer and the multiplication layer. Similarly, for each cross-modal interaction, the multihead attention layer performs the cross-attention on the text embeddings F_l and the single-level visual embeddings F_v^i, where the visual embeddings act as the query (Q) and the text embeddings serves as the key-value pair (K, V). The attention layer first derives the cross-modal similarity matrix $A_{vl}^i \in R^{H_iW_i\times L}$ by computing the similarity between each pixel embedding and each word embedding as follows:

$$A_{vl}^i = \text{Softmax}\left(\frac{F_v^i W^Q \cdot (F_l W^K)^T}{\sqrt{d_k}}\right) \tag{3}$$

where $W^Q and W^K \in R^{C\times d_k}$ are learnable linear projections. Then, the similarity matrix is used to aggregate the vision-related object information in the text embeddings followed by the multiplication with the input visual embeddings F_v^i to obtain the refined single-level visual feature $F_v^{i'}$

$$F_v^{i'} = A_{vl}^i F_l W^V \cdot F_v^i \tag{4}$$

where W^V is the learnable linear projection. To obtain positional information, we add fixed one-dimensional sinusoidal positional encoding to the text features before the cross-attention operation.

After the cross-modal interaction between each level of the multilevel visual features with the raw text features, we obtain the refined multilevel visual features F_v'.

D. Multimodal DETR

The multimodal Transformer aims to exploit the refined sentence feature and the refined multilevel visual features to produce the target-aware object embeddings, which are used to predict the object coordinate. Considering the importance of multiscale features in RS image target localization [39], [40], [41] and the efficiency of the model, here we adopt the Deformable-DETR [25] as the multimodal DETR and use the sentence feature F_l^s as the language queries of the decoder to retrieve and aggregate the referred object information from the visual features.

As shown in Fig. 2, the multimodal Transformer is equipped with an encoder-decoder structure, where the encoder builds the relationship among the key pixels and allows the model focusing on the key contents in visual features, and the decoder realizes the cross-modal interaction between the text and visual embeddings. As depicted in Fig. 4, the encoder is composed of four encoder layers, each of which contains a multiscale deformable self-attention (MS-DSA) module [25] and a fully connected feed-forward network (FFN). Similarly, the decoder also consists of four decoder layers, and each layer includes a self-attention (SA) module [24], a multiscale deformable cross-attention (MS-DCA) module [25] and a FFN. The multiscale deformable attention could be formulated as follows:

$$\begin{aligned}&\text{MS-DA}\left(\boldsymbol{z}_q, \hat{\boldsymbol{p}}_q, \{\boldsymbol{x}^l\}_{l=1}^L\right)\\ &= \sum_{m=1}^{M} \boldsymbol{W}_m \left[\sum_{l=1}^{L}\sum_{k=1}^{K} A_{mlqk} \cdot \boldsymbol{W}_m' \boldsymbol{x}^l\left(\phi_l\left(\hat{\boldsymbol{p}}_q\right) + \Delta \boldsymbol{p}_{mlqk}\right)\right]\end{aligned} \tag{5}$$

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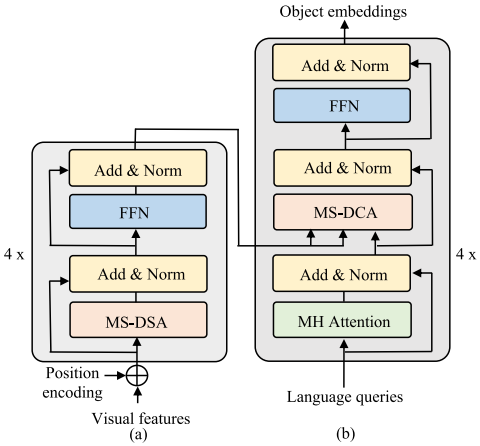


Fig. 4. Structure of the mulitmodal DETR. (a) Transformer encoder. (b) Transformer decoder. MS-DSA denotes the multiscale deformable self-attention module. MS-DCA denotes the multiscale deformable cross-attention module.

where $\boldsymbol{z}_q \in \mathbb{R}^C$ is the qth query element of the queries embeddings, $\hat{\boldsymbol{p}}_q$ is the normalized coordinates of the reference point for the qth query element, and $\{\boldsymbol{x}^l\}_{l=1}^L$ are the multiscale feature maps. m indexes the attention head, l indexes the input feature level, k indexes the sampling point, and K is the total sampled points in each level of feature. $\Delta \boldsymbol{p}_{mlqk} \in \mathbb{R}^2$ and $A_{mlqk} \in [0, 1]$ denote the sampling offset and attention weight of the kth sampling point in the lth feature level and the mth attention head, respectively. Both $\Delta \boldsymbol{p}_{mlqk}$ and A_{mlqk} are obtained via linear projection over the query embedding $\boldsymbol{z}_q$, and the scalar attention weight A_{mlqk} is normalized by $\sum_{l=1}^{L}\sum_{k=1}^{K} A_{mlqk} = 1$. $\boldsymbol{W}'_m$ is an identity matrix and $\phi_l(\cdot)$ is the function that rescales the normalized coordinates $\hat{\boldsymbol{p}}_q$ to the input feature map of the lth level.

As depicted in Fig. 4, the encoder takes as input the refined multiscale visual features with position encoding and output the multiscale visual features with the same resolutions after four encoder layers. In order to comprehensively learn the target representation in the decoder, we repeat the sentence feature $F_l^s \in \mathbb{R}^{1\times C}$ for N times to generate the language queries $Q_l \in \mathbb{R}^{N\times C}$ for each RS image. Subsequently, the language queries, the learnable reference point embeddings, and the encoder output are fed into the decoder. The language queries first go through the self-attention module and then interact with the multiscale visual features in the deformable cross-attention module to retrieve and aggregate target representation information. Finally, the Transformer decoder delivers the object embeddings for the target location prediction.

E. Object Function

As presented in Fig. 2, two task heads are built on top of the decoder to further transform the N object embeddings output from the decoder, i.e., class head and box head. The class head is a linear layer and predicts whether the predicted object is described by the language expression. The box head is a three-layer FFN with ReLU activation except for the last layer. It predicts the bounding box coordinate of the referred object. Both the two task heads generate N predictions for the input RS image.

During the training process, we adopt the bipartite matching strategy [1] to match the predicted and ground truth objects and accomplish the supervised training of the model. We denote the object predictions as $\hat{y} = \{\hat{y}_i\}_{i=1}^N$, and the ith prediction for the object is denoted as

$$\hat{y}_i = \left\{\hat{p}_i, \hat{b}_i\right\} \tag{6}$$

where $\hat{p}_i \in R^1$ is the probability score predicted by the class head. $\hat{b}_i \in R^4$ is the normalized coordinates that defines the center point as well as the height and width of the prediction box.

The ground truth could be represented as $y = \{c, b\}$, where c is an one-hot value that equals 1 when the ground truth object is visible in the RS image and 0 otherwise. b is the corresponding normalized box coordinates. To train the network, we first find the best prediction from all the N object predictions as the positive sample by minimizing the following matching loss:

$$\hat{y}_{\text{pos}} = \underset{\hat{y}_i \in \hat{y}}{\arg\min}\, \mathcal{L}_{\text{match}}\left(y, \hat{y}_i\right) \tag{7}$$

where

$$\mathcal{L}_{\text{match}}\left(y, \hat{y}_i\right) = \lambda_1 \mathcal{L}_{\text{cls}}\left(c, \hat{p}_i\right) + \lambda_2 \mathcal{L}_{L1}\left(b, \hat{b}_i\right) + \lambda_3 \mathcal{L}_G\left(b, \hat{b}_i\right) \tag{8}$$

here, $\mathcal{L}_{\text{cls}}$ is the focal loss [42] for classification task and $\mathcal{L}_{L1}$ and $\mathcal{L}_G$ are the L1 loss and GIoU loss [21], respectively, for the bounding box prediction task. We set $\lambda_1 = 2$, $\lambda_2 = 5$, and $\lambda_3 = 2$. Then, the whole model is optimized by minimizing the matching loss of the positive sample.

During inference, given an RS image and the language description, LQVG predicts N objects candidates that includes the class probability and object coordinate corresponding to the N queries and we select the object coordinate prediction with the highest class probability score as the final prediction of the RS image.

IV. Experiments

In this section, we conduct extensive experiments on two RSVG datasets to demonstrate the effectiveness of our proposed LQVG in terms of both quantitative and qualitative results.

A. Datasets

1) RSVG-HR: We construct the RSVG-HR dataset based on the very high-resolution RS images collected in RSVG dataset [7]. Since the language annotations in RSVG dataset heavily rely on the distance and area attributes of the objects which is incompatible with the common visual grounding methods, we rewrite the language annotations with new protocol that has more clear descriptions, i.e., **absolute position + relative position**. For example, as shown in Fig. 5(a), the language annotation of the fourth RS image is "a basketball court on the center of the far left and it is on the left of a swimming pool" where "the center of the far left" is the absolute position

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Fig. 5. Some visual examples of RSVG-HR and DIOR-RSVG datasets. (a) RSVG-HR dataset. (b) DIOR-RSVG datasets.

in the image and "the left of a swimming pool" is the relative position. RSVG-HR dataset has seven categories of objects and it consists of 2650 image–language pairs, where 2151 image–language pairs are used for training and the rest for test. The size of each RS image is 1024 × 1024 and the average length of the language descriptions is 19.6 words. The detailed information about RSVG-HR dataset can be found in Table I, and some visual examples are shown in Fig. 5(a).

2) DIOR-RSVG: DIOR-RSVG dataset is released by the Northwestern Polytechnical University in [8]. The dataset is built on the DIOR dataset and contains 20 object categories. DIOR-RSVG consists of 38 320 language expressions across 17 402 RS images, with 800 × 800 image size and 7.47 average expression length. Some visual examples of DIOR-RSVG dataset are shown in Fig. 5(b), and it can be observed that the language expressions in DIOR-RSVG dataset are somewhat simple and ambiguous. Moreover, since the dataset division of released version is different from the division claimed in the original paper, we evaluate and report the performance of our proposed LQVG and the comparison methods on the released version, where the training set, validation set, and test set have 26 991 (70%), 3829 (10%), and 7500 (20%) image–language pairs, respectively.

TABLE I
DETAIL OF THE RSVG-HR DATASET

No.	Class name	Training	Test
C01	Baseball field	443	81
C02	Basketball court	201	38
C03	Ground track field	163	50
C04	Roundabout	534	122
C05	Swimming pool	150	40
C06	Storage tank	163	50
C07	Tennis court	497	118
-	Total	2151	499

B. Evaluation Metrics

Following the evaluation metrics of MGVLF [8], we adopt the Pr@0.5, Pr@0.6, Pr@0.7, Pr@0.8, Pr@0.9, meanIoU, and cumIoU. Pr@0.5, Pr@0.6, Pr@0.7, Pr@0.8, and Pr@0.9 refer to the precisions that the intersection over union (IoU) of the predicted and the ground-truth bounding boxes are higher than the threshold at 0.5, 0.6, 0.7, 0.8, and 0.9, respectively. The meanIoU and cumIoU can be formulated as follows:

$$\text{meanIoU} = \frac{1}{M}\sum_{t} I_t / U_t \tag{9}$$

and

$$\text{cumIoU} = \left(\sum_{t} I_t\right) \Big/ \left(\sum_{t} U_t\right) \tag{10}$$

where t is the index of the image–language pairs and M represents the size of the dataset. I_t and U_t are the intersection and union area of predicted and ground-truth bounding boxes.

C. Implementation Details

1) Model Settings: We use the ResNet50 [38] pretrained on ImageNet as the image encoders. The input language sentence is first converted to token embeddings where the

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TABLE II
COMPARISON WITH STATE-OF-THE-ART METHODS ON THE DIOR-RSVG TEST SET. TIME MEANS THE RUNNING TIME OF PROCESSING EACH FRAME

Methods	Venue	Visual Enc.	Text Enc.	Params (M)	Pr@0.5	Pr@0.6	Pr@0.7	Pr@0.8	Pr@0.9	meanIoU	cumIoU	Time (s)
TransVG [31]	*ICCV'21*	ResNet50	BERT	149.7	66.18	61.18	53.58	41.11	18.05	57.25	69.48	0.8
LBYL-Net [43]	*CVPR'21*	DarkNet-53	BERT	163.8	67.47	61.84	54.21	39.89	16.24	58.33	70.62	1.3
QRNet [33]	*CVPR'22*	Swin-S	BERT	178.4	69.77	63.31	54.57	42.82	18.66	59.05	71.69	1.7
VLTVG [34]	*CVPR'22*	ResNet50	BERT	152.2	69.13	63.36	55.05	43.65	19.10	59.69	72.05	1.1
MGVLF [8]	*TGRS'23*	ResNet50	BERT	152.5	70.07	64.43	55.89	43.40	21.32	60.64	74.40	1.1
LQVG	-	ResNet50	BERT	166.3	**83.41**	**81.03**	**75.91**	**65.52**	**43.53**	**74.02**	**82.22**	1.4

TABLE III
COMPARISON WITH STATE-OF-THE-ART METHODS ON THE RSVG-HR TEST SET. TIME MEANS THE RUNNING TIME OF PROCESSING EACH FRAME

Methods	Venue	Visual Enc.	Text Enc.	Params (M)	Pr@0.5	Pr@0.6	Pr@0.7	Pr@0.8	Pr@0.9	meanIoU	cumIoU	Time (s)
TransVG [31]	*ICCV'21*	ResNet50	BERT	149.7	46.26	34.76	19.05	5.63	0.3	40.72	30.47	0.9
LBYL-Net [43]	*CVPR'21*	DarkNet-53	BERT	163.8	48.17	35.84	20.33	5.46	0.3	41.57	30.88	1.5
VLTVG [34]	*CVPR'22*	ResNet50	BERT	152.2	48.76	36.33	21.43	7.43	0.5	42.06	31.92	1.2
QRNet [33]	*CVPR'22*	Swin-S	BERT	178.4	50.35	37.76	21.77	7.92	0.5	42.87	32.74	1.9
MGVLF [8]	*TGRS'23*	ResNet50	BERT	152.5	50.70	38.88	23.45	8.22	0.6	43.11	33.23	1.2
LQVG	-	ResNet50	BERT	166.3	**87.37**	**82.57**	**72.34**	**51.10**	**14.43**	**71.55**	**70.06**	1.6

[CLS] token is inserted at the first place, and then the widely used $\text{BERT}_{\text{BASE}}$ model [10] initialized with the pretrained weights is employed as the text encoder to encode the tokens and generate the final text features. For the Multimodal DETR, we adopt four encoder layers and four decoder layers with the hidden dimension $C = 256$. The number of language queries $N = 10$.

2) Training Details: We implement our LQVG model using the Pytorch framework and train it in a distributed fashion on four NVIDIA 2080ti GPUs. For the DIOR-RSVG dataset, we train the model on the DIOR-RSVG training set with batch size of 2 per GPU for 70 epochs, where the learning rate is reduced by a factor of 0.1 at the 40th and 60th epoch, respectively. The size of each RS image is randomly resized to the range of [480, 560, 640, 720, 800]. The model is optimized using the AdamW optimizer with the initial learning rate of 1×10^{-5} for the text encoders and 1×10^{-4} for the rest parts. For the RSVG-HR dataset, we fine-tune the model trained on the DIOR-RSVG training set on the RSVG-HR training set with the same training setting as DIOR-RSVG dataset. Particularly, the size of each RS image in RSVG-HR training set is randomly resized to the range of [768, 840, 920, 1024]. Notably, the input RS image will not be resized during the inference process, but keeps its original size.

D. Comparison With State-of-the-Art Methods

To verify the superiority of the proposed LQVG, we compare it with several state-of-the-art methods of visual grounding for both natural image and RS image, including TransVG [31], VLTVG [34], and MGVLF [8], on the DIOR-RSVG test set and RSVG-HR test set. *The reported results of the comparison methods are reproduced using the official open source codes. The running time of each model is measured on the same platform.*

1) DIOR-RSVG: The comparison results on DIOR-RSVG test set are reported in Table II. It can be observed that our proposed LQVG sets a new state-of-the-art record in all metrics with competitive model parameters and running time. Especially, compared with the latest MGVLF methods, our LQVG significantly outperforms it across all the metrics, e.g., 13.34% higher in Pr@0.5, 22.12% higher in Pr@0.8, and 13.38% in meanIoU. MGVLF, VLTVG, QRNet, and TransVG all choose the architecture that relies on a learnable object query to learn contextual information of the referred object from the concatenated visual and text features via self-attention, while we propose the language query-based multimodal DETR in which the sentence feature is directly used as the language queries to retrieve and aggregate the object representation information from the multiscale visual features through the deformable cross-attention, and the experimental results demonstrate the superiority of the language query-based Transformer framework.

Fig. 6 presents the visual comparison between MGVLF and LQVG, and it can be seen that our LQVG could predict more accurate bounding boxes for the referred objects and reduces the false localizations. E.g., for the RS image with language expression of "the storage tank on the top," our LQVG accurately finds the referred object while MGVLF incorrectly predicts into the airplane.

2) RSVG-HR: We further evaluate the performance of LQVG and the comparison methods on RSVG-HR test set and the results are included in Table III. All the methods are first pretrained on DIOR-RSVG training set and then fine-tuned on the RSVG-HR training set. As shown in Table III, with the same image and text encoders, our LQVG outperforms the comparison methods in all the metrics by a large margin. For instance, our LQVG is 48.89% and 36.83% higher than MGVLF on Pr@0.7 and cumIoU, respectively. Particularly, MGVLF only has 0.6% on the Pr@0.9 while our LQVG can achieve 14.43% on this metrics. Some visual comparisons between LQVG and MGVLF are depicted in Fig. 7 and we can find that our LQVG can precisely identify the inconspicuous objects in the very high-resolution RS images while MGVLF fails. The quantitative and qualitative results on RSVG-HR

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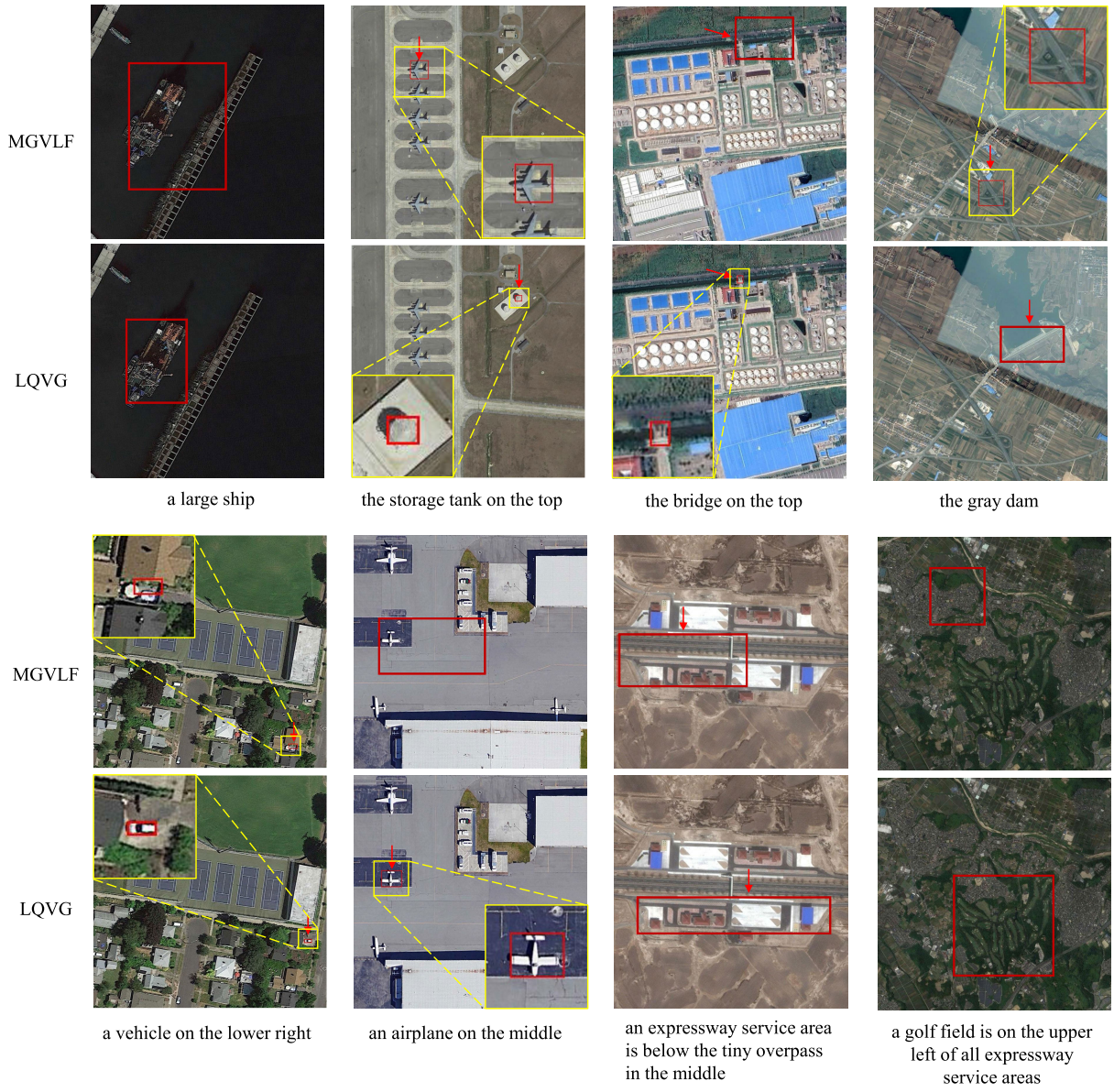


Fig. 6. Visual comparisons between MGVLF and our proposed LQVG on DIOR-RSVG dataset.

test set reveal that the previous RSVG methods may suffer unsatisfactory performance once the RS images have very high resolutions and the referred objects are hidden in a complex background. Our LQVG, however, is able to realize accurate target localization, demonstrating the superiority and effectiveness of our language query-based multimodal DETR framework.

E. Model Analysis

In this part, we conduct the ablation studies on the core components of LQVG and explore the influence of different hyperparameters on the model performance.

1) MSCMA Module: In order to comprehensively understand the impact of the MSCMA module on the performance of the LQVG model, we carry out full ablation studies on this module and present the results as well as the complexity information on Table IV. For the base LQVG model that without MSCMA module, i.e., the LVI and VLI submodules are not equipped, the model only reaches 62.99% and 60.17% meanIoU in the DIOR-RSVG and RSVG-HR datasets, respectively, which degrades severely compared to the full LQVG model. When LQVG model cooperates only with the LVI submodule, the meanIoU increases to 66.65% and 63.67% on the two datasets. The results indicate that the introduction of the LVI submodule brings 3.66% and 3.5% gains to the base model in meanIoU on the two datasets. Alternatively, when the LQVG model is only equipped with the VLI submodule, the meanIoU on the two datasets are 8.58% and 8.12% higher than the base model, respectively. Finally, when LQVG includes the full MSCMA module, the model obtains the best performance in all metrics. The results of this ablation study show that the refined multiscale visual features plays more important role in the cross-modal alignment module while the combination of

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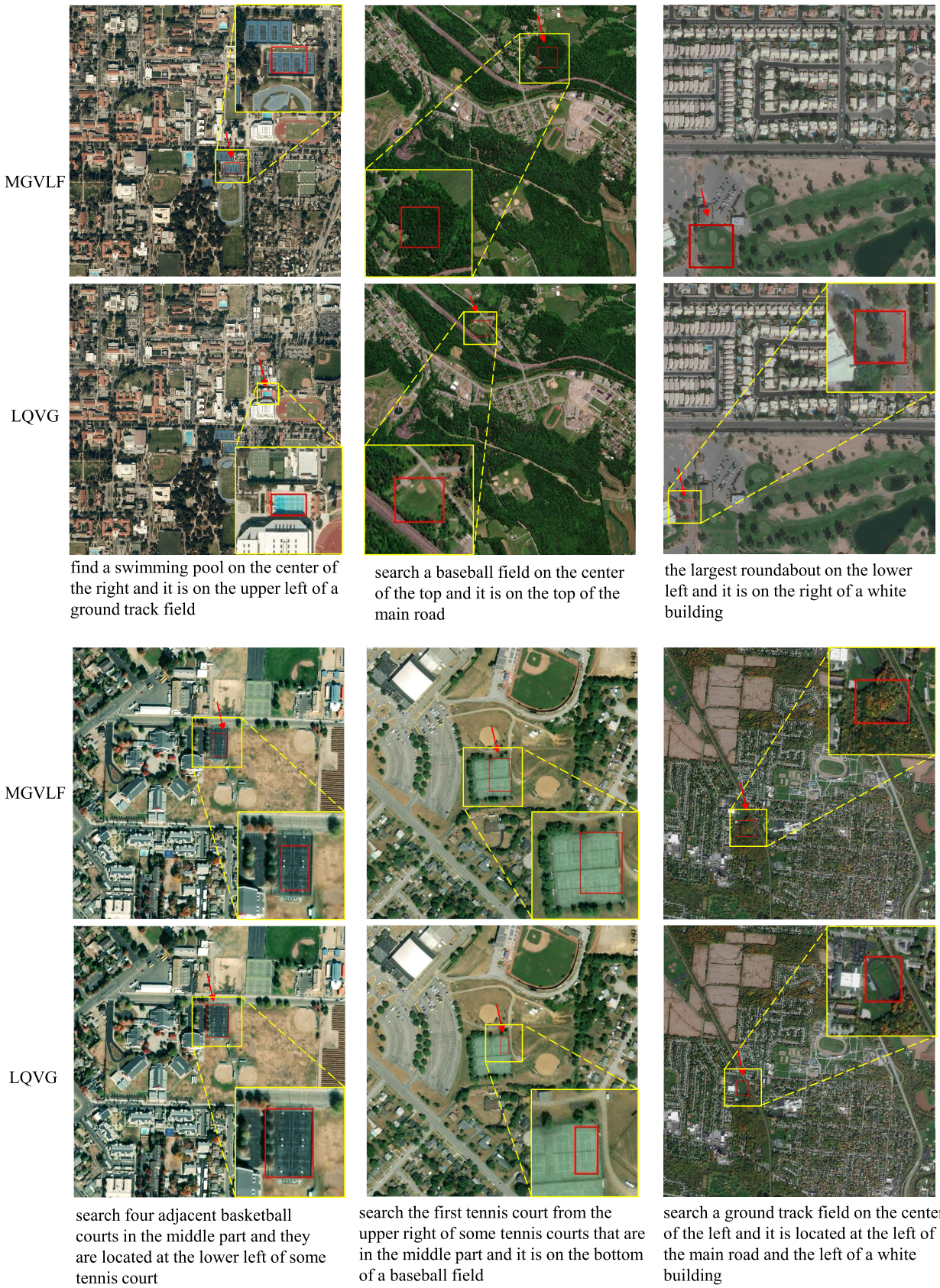


Fig. 7. Visual comparisons between MGVLF and our proposed LQVG on RSVG-HR dataset.

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TABLE IV
ABLATION STUDY ON THE TWO SUBMODULES OF MSCMA MODULE

Model	LVI	VLI	Params (M)	DIOR-RSVG Pr@0.7	DIOR-RSVG meanIoU	RSVG-HR Pr@0.7	RSVG-HR meanIoU	Time (s)
LQVG	×	×	165.9	64.89	62.99	61.86	60.17	1.594
LQVG	✓	×	166.1	67.77	66.65	64.33	63.67	1.616
LQVG	×	✓	166.1	74.04	71.57	68.14	68.29	1.603
LQVG	✓	✓	166.3	**75.91**	**74.02**	**72.34**	**71.55**	1.625

TABLE V
ABLATION STUDY OF INTERACTION STRATEGIES IN MSCMA MODULE

Interaction Strategies	DIOR-RSVG Pr@0.6	DIOR-RSVG Pr@0.7	DIOR-RSVG meanIoU	RSVG-HR Pr@0.6	RSVG-HR Pr@0.7	RSVG-HR meanIoU
Attention+Add	80.80	75.63	73.84	82.02	72.00	71.15
Attention+Add+FFN	80.86	75.27	73.90	82.16	72.08	71.23
Attention+multiply	**81.03**	**75.91**	**74.02**	**82.57**	**72.34**	**71.55**

both submodules leads to the best accuracy. Moreover, since the LVI and VLI submodules have the same structure, they have the same number of parameters, while the LVI submodule consumes more computation, resulting in longer running time.

2) Interaction Strategies in MSCMA Module: In the traditional attention mechanism, the attention module is always followed with the residual connection and the FFN. However, recent work [28] also proposes to apply the multiplication layer after the attention module. To figure out which strategy is better for the cross-modal alignment of our model, we assess the performance of the LQVG model that with different strategies. The results are reported in Table V. It can be observed that the interaction strategy of attention followed with a multiplication layer performs better than the other two strategies, i.e., attention module followed with addition operation and the regular attention module followed with FFN. Therefore, our LQVG employs the attention module with multiplication layer as the default setting.

3) Query Number: In our LQVG model, the language queries play the crucial role in learning the discriminative feature representation of the referred objects. Different from the previous approaches that only adopt one learnable object token in the multimodal fusion Transformer, we suggest using multiple language queries to learn target representation more comprehensively and generate multiple candidates for the best prediction. To explore the impact of the number of language queries on model performance, we evaluate the performance of LQVG with different query number and the results are presented in Table VI. As we can see, the performance of the LQVG model gradually improves as the number of queries grows, peaking at ten queries and then starting to degrade. The experimental results show that it is not necessary that the higher query number leads to better performance, and the higher query number may instead result in suboptimal accuracy due to inaccurate class probabilities. According to the empirical results, we set the query number $N = 10$ as default.

4) Multiscale Visual Features: In the cross-modal alignment module, multiscale features are used to interact with the text features, however, whether multiscale features are essential for the module and how many levels of multiscale visual features are adequate has not been explored. Therefore, we conduct the ablation study on the number of the visual feature scales in the MSCMA module to verify the multiscale features setting. The experimental results are included in Table VII. It can be observed that when we utilize the four-level multiscale visual features, LQVG model achieves the best performance on both datasets. In contrast, when only single-scale visual features are applied in the module, LQVG only gets 70.98% and 68.19% in meanIoU on both datasets, which are respectively 3.04% and 3.45% lower than best version. Moreover, we can also find that the model performance progressively improves as the number of multiscale feature levels increases. We attribute this phenomenon to the fact that multiscale visual features can interact more comprehensively with the text features, thereby generating more correlated cross-modal features and facilitating subsequent multiscale cross-modal interactions in the multimodal Transformer.

TABLE VI
MODEL ANALYSIS ON THE NUMBER OF LANGUAGE QUERY

Query Number	DIOR-RSVG Pr@0.6	DIOR-RSVG Pr@0.7	DIOR-RSVG meanIoU	RSVG-HR Pr@0.6	RSVG-HR Pr@0.7	RSVG-HR meanIoU
2	79.72	74.76	73.22	79.76	69.33	69.67
4	80.11	75.12	73.40	80.56	70.84	70.29
6	80.05	75.55	73.73	81.36	71.62	70.76
8	80.55	75.47	73.90	81.89	72.08	71.16
10	**81.03**	**75.91**	**74.02**	**82.57**	**72.34**	**71.55**
12	80.03	75.21	73,30	82.50	72.32	71.48

TABLE VII
MODEL ANALYSIS ON THE NUMBER OF VISUAL FEATURE SCALES IN MSCMA MODULE. 64, 32, 16, AND 8 INDICATE STRIDES OF FEATURES

Feature Scales	DIOR-RSVG Pr@0.6	DIOR-RSVG Pr@0.7	DIOR-RSVG meanIoU	RSVG-HR Pr@0.6	RSVG-HR Pr@0.7	RSVG-HR meanIoU
$\{64\}$	77.69	72.44	70.98	78.69	69.02	68.10
$\{64, 32\}$	78.41	73.41	71.94	79.56	69.73	68.62
$\{64, 32, 16\}$	80.44	74.61	73.60	81.27	70.88	70.74
$\{64, 32, 16, 8\}$	**81.03**	**75.91**	**74.02**	**82.57**	**72.34**	**71.55**

V. CONCLUSION

In this article, we propose a novel language query-based multimodal Transformer framework for visual grounding on RS images. Specifically, a MSCMA module is designed to enhance the semantic correlation between the visual and text features and produce the refined multiscale visual features and sentence feature. In the multimodal DETR, we use the sentence feature as the language queries to directly retrieve and aggregate the referred object information from the multiscale visual features and generate the object embeddings for the final localization prediction. Besides, a new RSVG dataset based on very high-resolution RS images with new annotation protocol is built to evaluate the performance of the existing and our proposed RSVG methods. The experimental results on two RSVG datasets demonstrate the superiority of our model over state-of-the-art RSVG approaches for RSVG task.

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In the future work, the multimodal RS foundation model will be considered to enhance the zero-shot learning ability of the model.

REFERENCES

[1] N. Carion, F. Massa, G. Synnaeve, N. Usunier, A. Kirillov, and S. Zagoruyko, "End-to-end object detection with transformers," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, 2020, pp. 213–229.

[2] L. Zhang and L. Zhang, "Artificial intelligence for remote sensing data analysis: A review of challenges and opportunities," *IEEE Geosci. Remote Sens. Mag.*, vol. 10, no. 2, pp. 270–294, Jun. 2022.

[3] W. Wang, J. Zhang, and Y. Cao, "Towards data-efficient detection transformers," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, 2022, pp. 88–105.

[4] X. Lu, B. Wang, X. Zheng, and X. Li, "Exploring models and data for remote sensing image caption generation," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 4, pp. 2183–2195, Apr. 2018.

[5] Y. Xu, T. Bai, W. Yu, S. Chang, P. M. Atkinson, and P. Ghamisi, "AI security for geoscience and remote sensing: Challenges and future trends," *IEEE Geosci. Remote Sens. Mag.*, vol. 11, no. 2, pp. 60–85, Jun. 2023.

[6] C. Yang, Z. Li, and L. Zhang, "Bootstrapping interactive image–text alignment for remote sensing image captioning," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, pp. 1–12, 2024.

[7] Y. Sun, S. Feng, X. Li, Y. Ye, J. Kang, and X. Huang, "Visual grounding in remote sensing images," in *Proc. 30th ACM Int. Conf. Multimedia (MM)*, 2022, pp. 404–412.

[8] Y. Zhan, Z. Xiong, and Y. Yuan, "RSVG: Exploring data and models for visual grounding on remote sensing data," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, pp. 1–13, 2023.

[9] K. Li, G. Wan, G. Cheng, L. Meng, and J. Han, "Object detection in optical remote sensing images: A survey and a new benchmark," *ISPRS J. Photogramm. Remote Sens.*, vol. 159, pp. 296–307, Jan. 2020.

[10] J. D. M.-W. C. Kenton and L. K. Toutanova, "BERT: Pre-training of deep bidirectional transformers for language understanding," in *Proc. NAACL-HLT*, 2019, pp. 4171–4186.

[11] R. Hu, H. Xu, M. Rohrbach, J. Feng, K. Saenko, and T. Darrell, "Natural language object retrieval," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 4555–4564.

[12] L. Yu et al., "MAttNet: Modular attention network for referring expression comprehension," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2018, pp. 1307–1315.

[13] M. Bajaj, L. Wang, and L. Sigal, "G3raphGround: Graph-based language grounding," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 4281–4290.

[14] X. Lu, Y. Zhang, Y. Yuan, and Y. Feng, "Gated and axis-concentrated localization network for remote sensing object detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 1, pp. 179–192, Jan. 2020.

[15] S. Yang, G. Li, and Y. Yu, "Graph-structured referring expression reasoning in the wild," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 9952–9961.

[16] H. Zhang, Y. Niu, and S.-F. Chang, "Grounding referring expressions in images by variational context," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2018, pp. 4158–4166.

[17] S. Yang, G. Li, and Y. Yu, "Relationship-embedded representation learning for grounding referring expressions," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 43, no. 8, pp. 2765–2779, Aug. 2021.

[18] A. Sadhu, K. Chen, and R. Nevatia, "Zero-shot grounding of objects from natural language queries," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 4694–4703.

[19] Y. Liao, A. Zhang, Z. Chen, T. Hui, and S. Liu, "Progressive language-customized visual feature learning for one-stage visual grounding," *IEEE Trans. Image Process.*, vol. 31, pp. 4266–4277, 2022.

[20] Z. Yang, B. Gong, L. Wang, W. Huang, D. Yu, and J. Luo, "A fast and accurate one-stage approach to visual grounding," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 4683–4693.

[21] H. Rezatofighi, N. Tsoi, J. Gwak, A. Sadeghian, I. Reid, and S. Savarese, "Generalized intersection over union: A metric and a loss for bounding box regression," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 658–666.

[22] Y. Liao et al., "A real-time cross-modality correlation filtering method for referring expression comprehension," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 10880–10889.

[23] Z. Yang, T. Chen, L. Wang, and J. Luo, "Improving one-stage visual grounding by recursive sub-query construction," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*. Springer, 2020, pp. 387–404.

[24] A. Vaswani et al., "Attention is all you need," in *Proc. Adv. Neural Inform. Process. Syst.*, 2017, pp. 5998–6008.

[25] X. Zhu, W. Su, L. Lu, B. Li, X. Wang, and J. Dai, "Deformable DETR: Deformable transformers for end-to-end object detection," 2020, *arXiv:2010.04159*.

[26] M. Lan, J. Zhang, L. Zhang, and D. Tao, "Learning to learn better for video object segmentation," in *Proc. AAAI Conf. Artif. Intell. (AAAI)*, 2023, pp. 1205–1212.

[27] L. Zhang, M. Lan, J. Zhang, and D. Tao, "Stagewise unsupervised domain adaptation with adversarial self-training for road segmentation of remote-sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–13, 2021.

[28] Z. Yang, J. Wang, Y. Tang, K. Chen, H. Zhao, and P. H. S. Torr, "LAVT: Language-aware vision transformer for referring image segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 18155–18165.

[29] Y. Du, Z. Fu, Q. Liu, and Y. Wang, "Visual grounding with transformers," in *Proc. IEEE Int. Conf. Multimedia Expo (ICME)*, Jul. 2022, pp. 1–6.

[30] M. Sun, W. Suo, P. Wang, Y. Zhang, and Q. Wu, "A proposal-free one-stage framework for referring expression comprehension and generation via dense cross-attention," *IEEE Trans. Multimedia*, vol. 25, pp. 2446–2458, 2022.

[31] J. Deng, Z. Yang, T. Chen, W. Zhou, and H. Li, "TransVG: End-to-end visual grounding with transformers," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 1769–1779.

[32] J. Deng et al., "TransVG++: End-to-end visual grounding with language conditioned vision transformer," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 45, no. 11, pp. 13636–13652, Nov. 2023.

[33] J. Ye et al., "Shifting more attention to visual backbone: Query-modulated refinement networks for end-to-end visual grounding," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 15502–15512.

[34] L. Yang, Y. Xu, C. Yuan, W. Liu, B. Li, and W. Hu, "Improving visual grounding with visual-linguistic verification and iterative reasoning," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 9499–9508.

[35] C. Liu, R. Zhao, H. Chen, Z. Zou, and Z. Shi, "Remote sensing image change captioning with dual-branch transformers: A new method and a large scale dataset," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–20, 2022.

[36] C. Liu, R. Zhao, J. Chen, Z. Qi, Z. Zou, and Z. Shi, "A decoupling paradigm with prompt learning for remote sensing image change captioning," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, pp. 1–18, 2023.

[37] Y. Xu, W. Yu, P. Ghamisi, M. Kopp, and S. Hochreiter, "Txt2Img-MHN: Remote sensing image generation from text using modern Hopfield networks," *IEEE Trans. Image Process.*, vol. 32, pp. 5737–5750, 2023.

[38] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 770–778.

[39] G. Cheng et al., "Anchor-free oriented proposal generator for object detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5625411.

[40] J. Yang, B. Du, Y. Xu, and L. Zhang, "Can spectral information work while extracting spatial distribution?—An online spectral information compensation network for HSI classification," *IEEE Trans. Image Process.*, vol. 32, pp. 2360–2373, 2023.

[41] R. Xu, X.-M. Dong, W. Li, J. Peng, W. Sun, and Y. Xu, "DBCT-Net: Double branch convolution-transformer network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, pp. 1–15, 2024.

[42] T.-Y. Lin, P. Goyal, R. Girshick, K. He, and P. Dollár, "Focal loss for dense object detection," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Oct. 2017, pp. 2980–2988.

[43] B. Huang, D. Lian, W. Luo, and S. Gao, "Look before you leap: Learning landmark features for one-stage visual grounding," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 16888–16897.

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Meng Lan received the B.S. and Ph.D. degrees from the School of Computer Science, Wuhan University, Wuhan, China, in 2018 and 2023, respectively.

He is currently a Post-Doctoral Researcher with the Department of Electronic and Computer Engineering, The Hong Kong University of Science and Technology, Hong Kong, China. His research interests include deep learning and computer vision.



Fu Rong received the B.E. degree (Hons.) in computer science and technology from the School of Computer Science, Wuhan University, Wuhan, China, in 2024, where he is currently pursuing the M.S. degree in computer science and technology.

His research interests include deep learning and computer vision.



Hongzan Jiao (Member, IEEE) received the Ph.D. degree in photogrammetry and remote sensing from Wuhan University, Wuhan, China, in 2013.

He is an Associate Professor at the School of Urban Design, Wuhan University. His research interests include urban remote sensing, urban big data mining, urban computing, pattern recognition, and artificial intelligence, including genetic algorithm and deep learning network.



Zhi Gao (Member, IEEE) received the B.Eng. and Ph.D. degrees from Wuhan University, Wuhan, China, in 2002 and 2007, respectively.

In 2008, he joined the Interactive and Digital Media Institute, National University of Singapore (NUS), Singapore, as a Research Fellow (A) and the Project Manager. In 2014, he joined the Temasek Laboratories, NUS (TL@NUS), Singapore, as a Research Scientist (A) and the Principal Investigator. He is currently working as a Full Professor with the School of Remote Sensing and Information Engineering, Wuhan University. He has published more than 70 research articles on top journals and conferences. His research interests include computer vision, machine learning, and remote sensing and their applications. In particular, he has strong interests in vision for intelligent systems and intelligent system-based vision.

Dr. Gao serves as an Associate Editor for *Unmanned Systems journal*.



Lefei Zhang (Senior Member, IEEE) received the B.S. and Ph.D. degrees from Wuhan University, Wuhan, China, in 2008 and 2013, respectively.

He was a Big Data Institute Visitor with the Department of Statistical Science, University College London, London, U.K., and a Hong Kong Scholar with the Department of Computing, The Hong Kong Polytechnic University, Hong Kong, China. He is a Professor with the School of Computer Science, Wuhan University, and with the Hubei Luojia Laboratory, Wuhan. His research interests include pattern recognition, image processing, and remote sensing.

Dr. Zhang serves as a Topical Associate Editor for IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING and an Associate Editor for IEEE GEOSCIENCE AND REMOTE SENSING LETTERS.

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